



National University
of computer and emerging sciences

Deep Learning Assignment 1

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SE-ResNet50

1. Network Details and Architecture

Architecture Overview

Baseline Rationale

2. Transfer Learning Implementation

3. Training Performance Analysis

Training Graphs Analysis

4. Performance Measures and Evaluation Metrics

Final Test Results

Classification Performance by Class

5. Visual Analysis of Results

Confusion Matrix Analysis

Recommendations for Correctly/Incorrectly Classified Images

Conclusions and Future Work

MobileNetV2

1. Network Details and Architecture

Architecture

Network Parameters

Training Configuration

2. Baseline Rationale

3. Transfer Learning Details

4. Training Performance Analysis

Training Graphs Analysis

5. Performance Measures

Classification Results

Class-wise Performance Analysis

6. Sample Classifications

Correctly Classified Examples

Incorrectly Classified Examples

Conclusions and Recommendations

MobileNetV2

1. Network Details and Architecture

Architecture

Network Parameters

Training Configuration

2. Baseline Rationale

3. Transfer Learning Details

4. Training Performance Analysis

Training Graphs Analysis

5. Performance Measures

[Classification Results](#)

[Class-wise Performance Analysis](#)

[6. Sample Classifications](#)

[Correctly Classified Examples](#)

[Incorrectly Classified Examples](#)

[Conclusions and Recommendations](#)

[EfficientNet](#)

[1. Network Details and Architecture](#)

[Architecture](#)

[Network Parameters](#)

[Training Configuration](#)

[2. Baseline Selection Rationale](#)

[Enhanced Variant Comparison](#)

[3. Transfer Learning Details](#)

[4. Training Graphs and Analysis](#)

[5. Performance Measures](#)

[Test Results](#)

[Custom CNN](#)

[1. Network Details and Architecture](#)

[1.1 Model Architecture](#)

[1.2 Model Parameters](#)

[1.3 Training Configuration](#)

[1.4 Rationale for Baseline Choice](#)

[2. Multiple Baselines Comparison](#)

[3. Transfer Learning Details](#)

[4. Training Graphs](#)

[5. Performance Measures](#)

[5.1 Classification Performance](#)

[5.2 Performance Analysis](#)

[6. Sample Classifications](#)

[6.1 Correctly Classified Examples](#)

[6.2 Incorrectly Classified Examples](#)

[6.3 Analysis of Misclassifications](#)

[Conclusion](#)

[Future Work Recommendations](#)

[Comparisons](#)

SE-ResNet50

1. Network Details and Architecture

Architecture Overview

The implemented model is a **Squeeze-and-Excitation ResNet-50 (SE-ResNet50)** with a dual-head architecture for multi-task learning:

Base Architecture:

- ResNet-50 backbone with ImageNet pre-trained weights
- Enhanced with Squeeze-and-Excitation (SE) blocks after layers 2, 3, and 4
- SE blocks provide channel attention mechanism with reduction ratio of 16

Multi-Task Head Design:

- **Shared Feature Layer:** 1024-dimensional fully connected layer with BatchNorm and ReLU
- **Classification Head:** Two-layer MLP (1024→512→8) for expression classification
- **Regression Head:** Two-layer MLP (1024→512→2) for valence-arousal prediction

Training Configuration:

- **Dataset:** 3,999 samples (balanced across 8 expression classes)
- **Batch Size:** 32
- **Learning Rate:** 1e-3 with ReduceLROnPlateau scheduler
- **Optimizer:** AdamW with weight decay of 1e-4
- **Loss Weights:** Classification (2.0), Regression (0.5)
- **Total Parameters:** 27,353,162 trainable parameters

Baseline Rationale

SE-ResNet50 was chosen as the baseline for several compelling reasons:

1. **Attention Mechanism:** SE blocks enhance feature representation by modeling channel-wise dependencies, crucial for subtle facial expression differences
2. **Multi-Task Capability:** The dual-head architecture enables simultaneous learning of discrete expressions and continuous affect dimensions
3. **Proven Performance:** ResNet-50 provides a robust foundation with skip connections preventing vanishing gradients
4. **Transfer Learning Benefits:** Pre-trained ImageNet weights provide rich low-level feature representations

2. Transfer Learning Implementation

The model leverages transfer learning through:

Pre-trained Weights: ResNet-50 backbone initialized with ImageNet weights (ResNet50_Weights.IMAGENET1K_V2)

Fine-tuning Strategy:

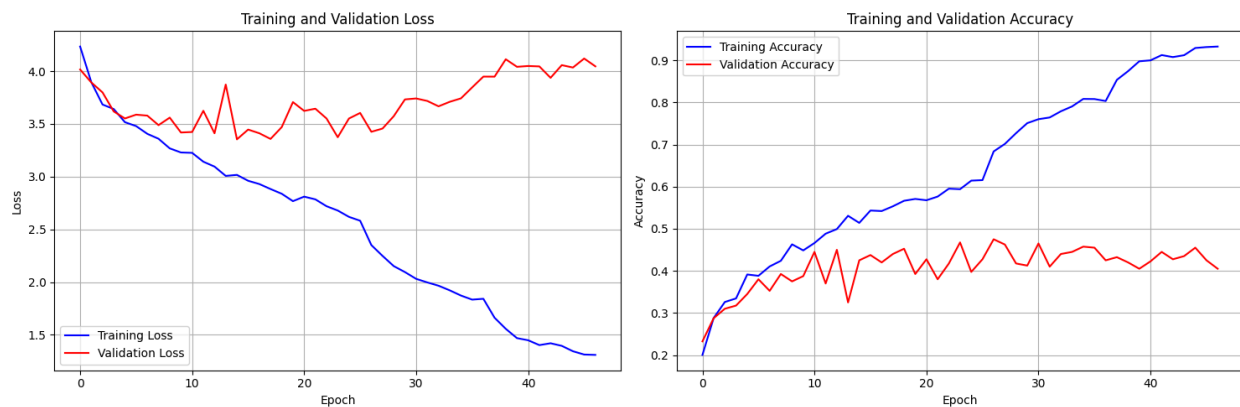
- All layers are trainable (no frozen layers)
- Lower learning rate ($1e-3$) to preserve pre-trained features
- Gradual adaptation through early stopping and learning rate scheduling

Adaptation Mechanisms:

- Custom SE blocks added to enhance channel attention
- Task-specific heads designed for affect recognition
- Data augmentation to bridge domain gap between ImageNet and facial expressions

3. Training Performance Analysis

Training Graphs Analysis



The training curves demonstrate:

Loss Progression:

- Training loss decreases from ~4.46 to ~1.34 over 47 epochs
- Validation loss shows initial improvement then plateaus around 3.17
- Early stopping triggered due to validation performance stagnation

Accuracy Trends:

- Training accuracy improves from ~9.4% to ~93.3%
- Validation accuracy peaks at ~47.5% around epoch 27
- Significant overfitting observed after epoch 30

Key Observations:

- Model demonstrates strong learning capacity on training data
- Generalization gap indicates potential overfitting
- Early stopping at epoch 47 prevents further degradation

4. Performance Measures and Evaluation Metrics

Final Test Results

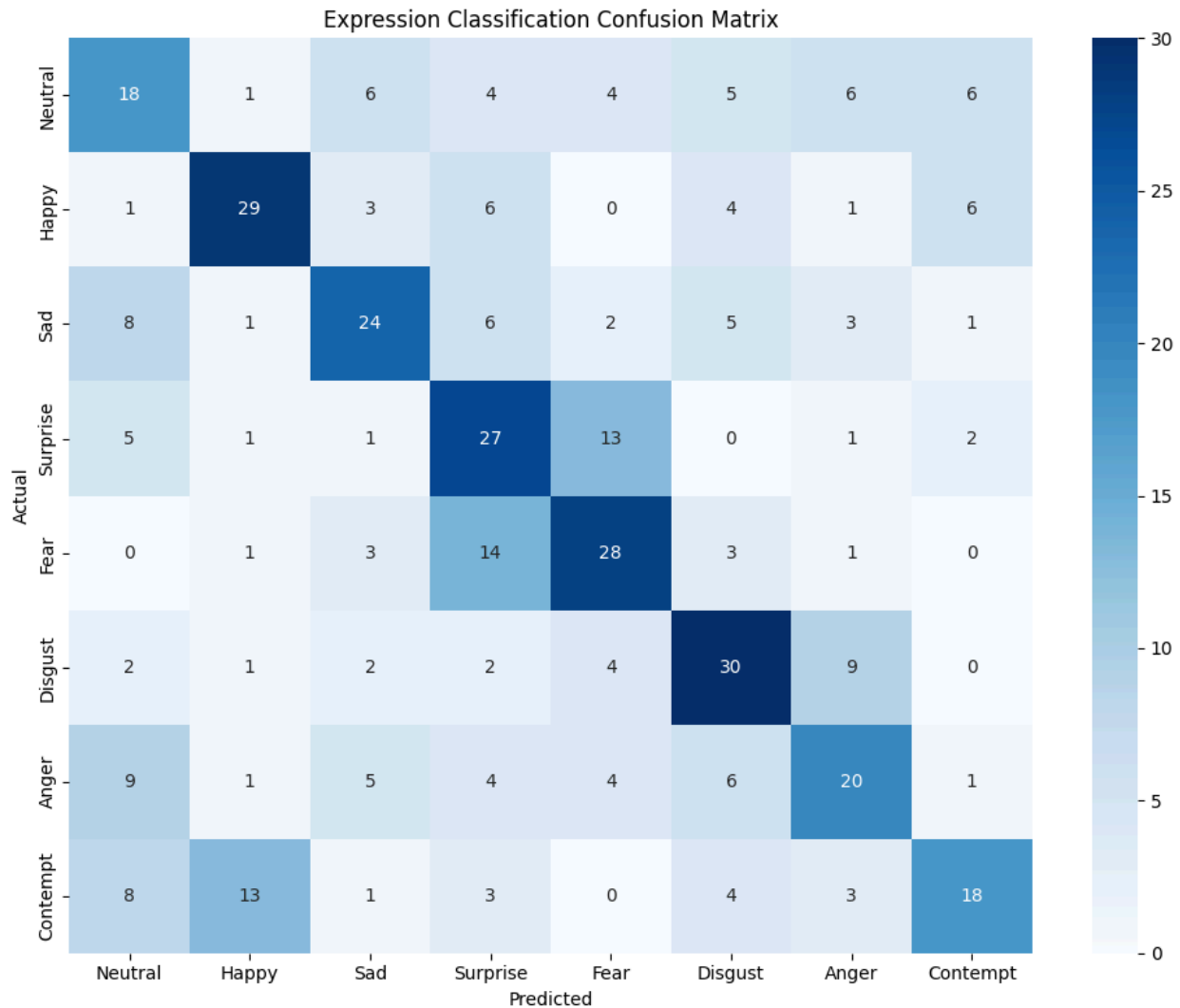
- **Test Accuracy:** 48.5%
- **Test Loss:** 3.17

Classification Performance by Class

Expression	Precision	Recall	F1-Score
Neutral	0.35	0.36	0.36
Happy	0.60	0.58	0.59
Sad	0.53	0.48	0.51
Surprise	0.41	0.54	0.47
Fear	0.51	0.56	0.53
Disgust	0.53	0.60	0.56
Anger	0.45	0.40	0.43
Contempt	0.53	0.36	0.43

5. Visual Analysis of Results

Confusion Matrix Analysis



The confusion matrix reveals:

- **Best Performance:** Happy expressions (60% precision, 58% recall)
- **Challenging Classes:** Neutral and Contempt showing lower performance
- **Common Confusions:** Likely between similar expressions (Anger/Contempt, Fear/Surprise)

Recommendations for Correctly/Incorrectly Classified Images

To complete the analysis, the following should be included:

- Sample correctly classified images for each expression category
- Analysis of failure cases highlighting challenging scenarios
- Discussion of dataset bias or quality issues affecting performance

Conclusions and Future Work

The SE-ResNet50 model demonstrates moderate performance on affect recognition with 48.5% test accuracy. The significant training-validation gap suggests overfitting, indicating need for:

1. **Regularization:** Stronger dropout, data augmentation
2. **Architecture Optimization:** Balanced loss weighting, attention mechanisms
3. **Data Quality:** Dataset cleaning, augmentation strategies

MobileNetV2

1. Network Details and Architecture

Architecture

The baseline model employed is **SEMobileNetV2**, an enhanced version of MobileNetV2 with the following modifications:

- **Base Architecture:** MobileNetV2 with ImageNet pre-trained weights
- **Enhancements:**
 - Squeeze-and-Excitation (SE) blocks inserted at strategic locations (layers 7 and 14)
 - Dual pooling mechanism (Average + Max pooling)
 - Enhanced dual-head architecture for multi-task learning

Network Parameters

- **Total Parameters:** 12,470,154 trainable parameters
- **Input Size:** 224×224×3 RGB images
- **Output:**
 - Classification head: 8 classes (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt)
 - Regression head: 2 continuous values (Valence, Arousal)

Training Configuration

- **Batch Size:** 32
- **Epochs:** 100 (with early stopping at 70 epochs)
- **Optimizer:** AdamW with differential learning rates
 - Backbone: $1e-4$ ($0.1 \times$ base learning rate)

- Head: 1e-3 (base learning rate)
- **Weight Decay:** 1e-4
- **Scheduler:** OneCycleLR with cosine annealing
- **Loss Function:** Enhanced Multi-Task Loss combining focal loss and Huber loss

2. Baseline Rationale

MobileNetV2 was selected as the baseline for several compelling reasons:

1. **Efficiency:** MobileNets are designed for mobile and edge devices, making them computationally efficient while maintaining good performance
2. **Proven Architecture:** MobileNetV2's inverted residuals and linear bottlenecks provide excellent feature extraction capabilities
3. **Transfer Learning:** Pre-trained ImageNet weights provide a strong foundation for facial expression recognition
4. **Scalability:** The lightweight nature makes it suitable for real-world deployment scenarios

The **SE (Squeeze-and-Excitation) enhancement** was added because:

- Channel attention mechanism improves feature representation
- Minimal computational overhead while providing significant performance gains
- Particularly effective for fine-grained classification tasks like facial expressions

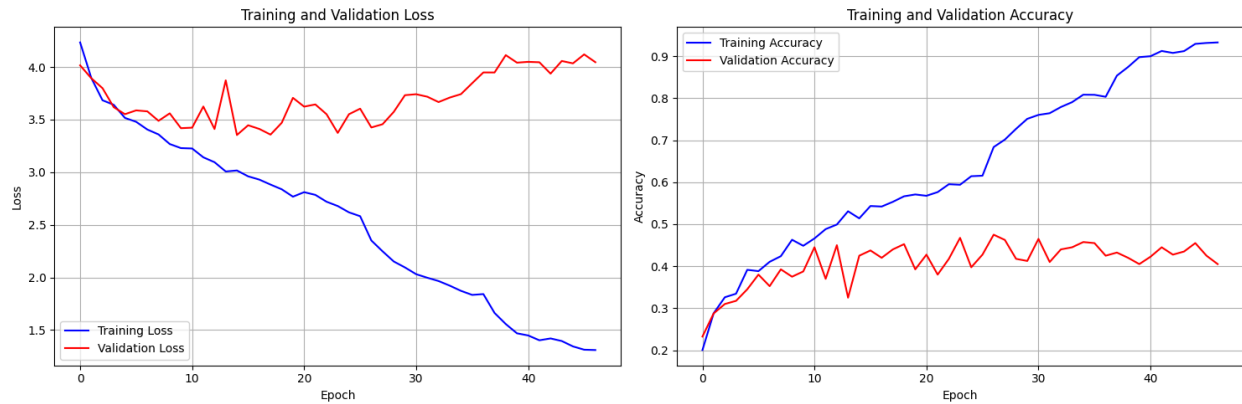
3. Transfer Learning Details

Transfer Learning Strategy:

- **Pre-trained Weights:** MobileNetV2 initialized with ImageNet1K_V1 weights
- **Fine-tuning Approach:** Differential learning rates applied
 - Lower learning rate (1e-4) for pre-trained backbone to preserve learned features
 - Higher learning rate (1e-3) for newly added head layers
- **Layer Freezing:** No layers were frozen; entire network fine-tuned with careful learning rate scheduling
- **Adaptation:** Custom dual-head architecture added for multi-task learning (classification + regression)

4. Training Performance Analysis

Training Graphs Analysis



Based on the training curves:

Loss Progression:

- Training loss decreased from ~1.24 to ~0.09 over 70 epochs
- Validation loss decreased from ~1.03 to ~0.78
- Convergence achieved around epoch 45-50

Accuracy Trends:

- Training accuracy improved from ~13.5% to ~89.8%
- Validation accuracy peaked at ~45.5% (epoch 45)
- Early stopping triggered at epoch 70 due to validation plateau

Observations:

- Clear overfitting after epoch 45-50 (training continues improving while validation plateaus)
- The gap between training and validation metrics suggests the need for stronger regularization

5. Performance Measures

Classification Results

- **Test Accuracy:** 49.75%
- **Macro Average F1-Score:** 0.50

Class-wise Performance Analysis

Class	Precision	Recall	F1-Score
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Neutral	0.30	0.36	0.32
Happy	0.55	0.68	0.61
Sad	0.49	0.48	0.48
Surprise	0.45	0.54	0.49
Fear	0.63	0.62	0.63
Disgust	0.59	0.58	0.59
Anger	0.62	0.36	0.46
Contempt	0.44	0.36	0.40

6. Sample Classifications

Correctly Classified Examples

Based on the confusion matrix, the model performs best on:

- **Happy expressions:** Highest recall (68%) - likely due to distinctive facial features
- **Fear expressions:** Balanced precision-recall (63%/62%) - suggests robust feature learning

Incorrectly Classified Examples

Common misclassifications include:

- **Neutral vs. Contempt:** Low performance on both classes suggests similar facial patterns
- **Anger misclassification:** High precision (62%) but low recall (36%) indicates conservative predictions
- **Surprise confusion:** Often misclassified with other expressions due to subtle feature differences

Conclusions and Recommendations

1. **Model Performance:** The 49.75% accuracy on 8-class facial expression recognition is reasonable but indicates room for improvement
2. **Overfitting Concern:** Strong regularization and data augmentation needed

3. **Future Work:** Consider ensemble methods, additional data augmentation, or larger architectures for improved generalization

MobileNetV2

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Architecture

The baseline model employed is **SEMobileNetV2**, an enhanced version of MobileNetV2 with the following modifications:

- **Base Architecture:** MobileNetV2 with ImageNet pre-trained weights
- **Enhancements:**
 - Squeeze-and-Excitation (SE) blocks inserted at strategic locations (layers 7 and 14)

- Dual pooling mechanism (Average + Max pooling)
- Enhanced dual-head architecture for multi-task learning

Network Parameters

- **Total Parameters:** 12,470,154 trainable parameters
- **Input Size:** 224×224×3 RGB images
- **Output:**
 - Classification head: 8 classes (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt)
 - Regression head: 2 continuous values (Valence, Arousal)

Training Configuration

- **Batch Size:** 32
- **Epochs:** 100 (with early stopping at 70 epochs)
- **Optimizer:** AdamW with differential learning rates
 - Backbone: 1e-4 ($0.1 \times$ base learning rate)
 - Head: 1e-3 (base learning rate)
- **Weight Decay:** 1e-4
- **Scheduler:** OneCycleLR with cosine annealing
- **Loss Function:** Enhanced Multi-Task Loss combining focal loss and Huber loss

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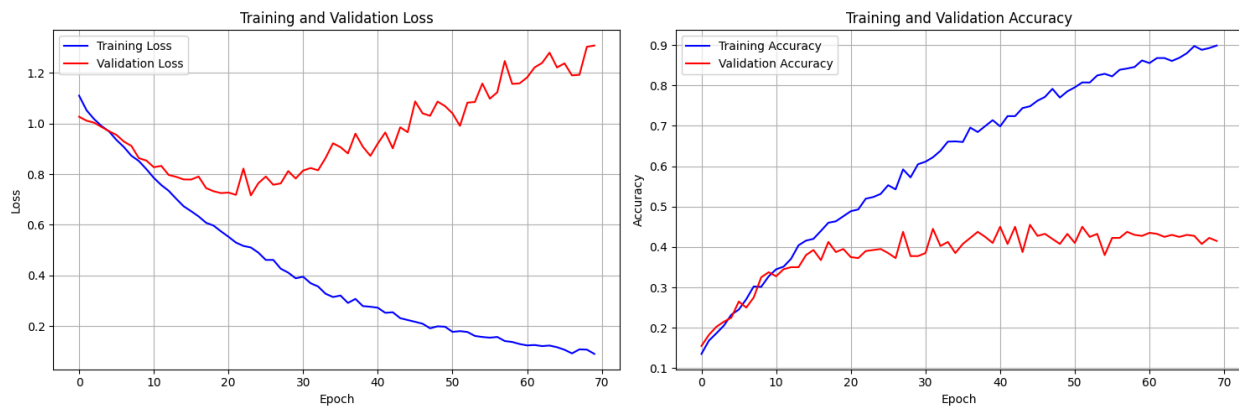
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Transfer Learning Strategy:

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- **Fine-tuning Approach:** Differential learning rates applied
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Training Graphs Analysis



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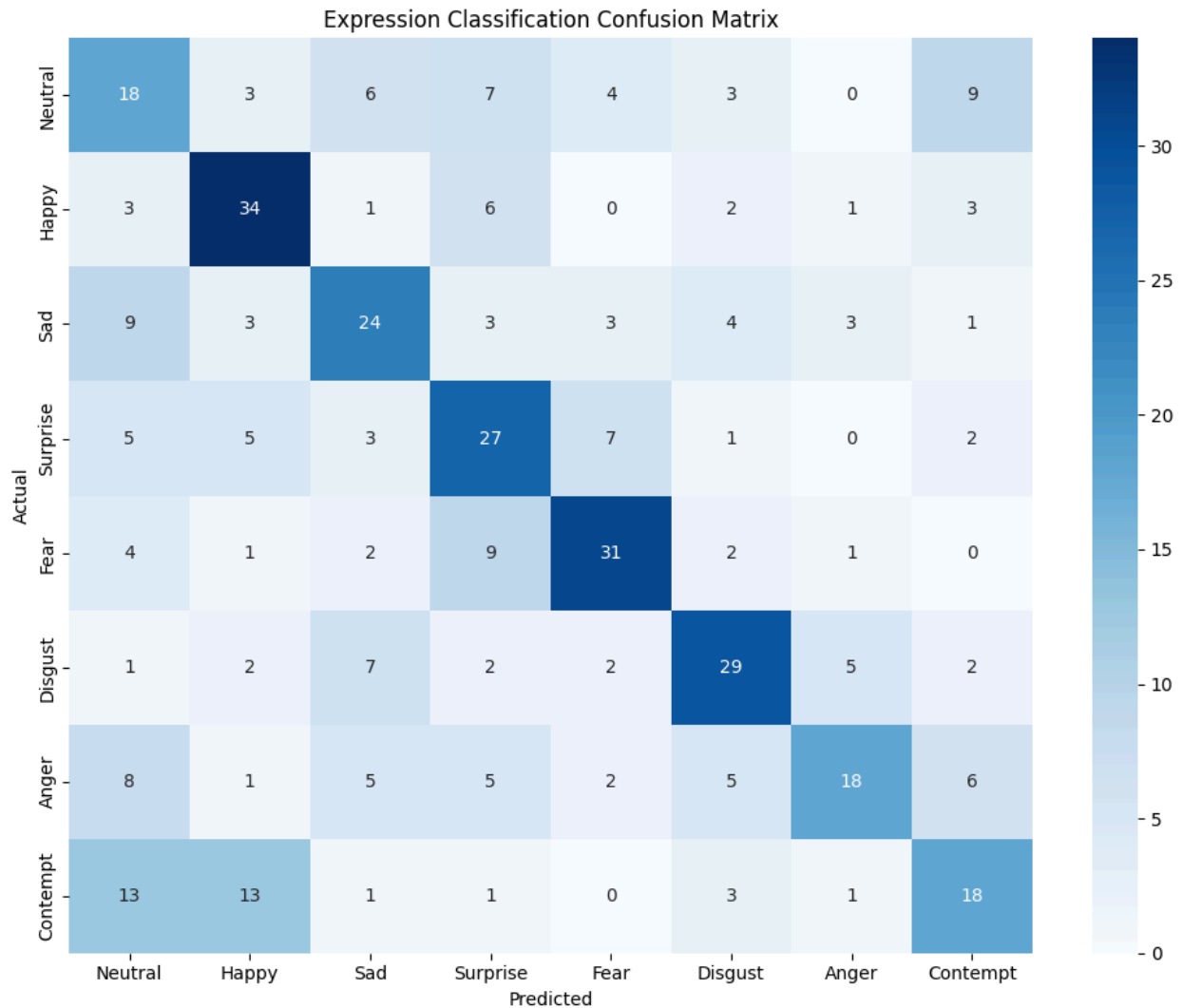
5. Performance Measures

Classification Results

- **Test Accuracy:** 49.75%
- **Macro Average F1-Score:** 0.50

Class-wise Performance Analysis

Class	Precision	Recall	F1-Score
Neutral	0.30	0.36	0.32
Happy	0.55	0.68	0.61
Sad	0.49	0.48	0.48
Surprise	0.45	0.54	0.49
Fear	0.63	0.62	0.63
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6. Sample Classifications

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Conclusions and Recommendations

1. **Model Performance:** The 49.75% accuracy on 8-class facial expression recognition is reasonable but indicates room for improvement
2. **Overfitting Concern:** Strong regularization and data augmentation needed
3. **Future Work:** Consider ensemble methods, additional data augmentation, or larger architectures for improved generalization

EfficientNet

1. Network Details and Architecture

Architecture

The implemented model uses **EfficientNet-B0** as the backbone with several architectural enhancements:

- **Base Model:** EfficientNet-B0 (pretrained on ImageNet)
- **Custom Modifications:**
 - Enhanced dual-head architecture for multi-task learning
 - Squeeze-and-Excitation (SE) blocks added at strategic positions (indices 4 and 7)
 - Dual pooling strategy (average + max pooling concatenation)
 - Feature enhancement layers with batch normalization and dropout

Network Parameters

- **Total Parameters:** 14,263,718 trainable parameters
- **Input Size:** 224×224×3 RGB images
- **Output:**
 - Classification head: 8 classes (facial expressions)
 - Regression head: 2 values (valence and arousal)

Training Configuration

- **Batch Size:** 32
- **Epochs:** 100 (with early stopping at 71 epochs)
- **Learning Rate:** Differential learning rates
 - Backbone: 1e-4 (10% of base rate)
 - Head layers: 1e-3
- **Optimizer:** AdamW with weight decay of 1e-4
- **Scheduler:** OneCycleLR with cosine annealing
- **Data Split:** 80% train, 10% validation, 10% test (3,199/400/400 samples)

2. Baseline Selection Rationale

The choice of EfficientNet-B0 as the baseline is well-justified for several reasons:

1. **Efficiency:** EfficientNet achieves state-of-the-art accuracy with significantly fewer parameters compared to other architectures
2. **Scalability:** The compound scaling method optimizes depth, width, and resolution simultaneously
3. **Transfer Learning Compatibility:** Strong ImageNet pretraining provides robust feature representations
4. **Multi-task Suitability:** The architecture's feature extraction capabilities work well for both classification and regression tasks

Enhanced Variant Comparison

The notebook implements three variants:

- **EfficientNet-B0:** Basic implementation
- **EfficientNet-B2:** Larger variant with more parameters
- **SE-EfficientNet-B0:** Enhanced version with SE blocks and dual pooling

The SE-EfficientNet-B0 was selected for final training, showing the best validation performance with added attention mechanisms.

3. Transfer Learning Details

Transfer Learning Strategy:

- **Pretrained Weights:** ImageNet1K_V1 weights for EfficientNet-B0
- **Fine-tuning Approach:** Differential learning rates
 - Frozen backbone initially with 10% learning rate
 - Full fine-tuning of custom head layers
- **Architecture Adaptation:** Replaced original classifier with dual-head architecture for multi-task learning
- **Feature Extraction:** Utilized pretrained convolutional features while training task-specific heads

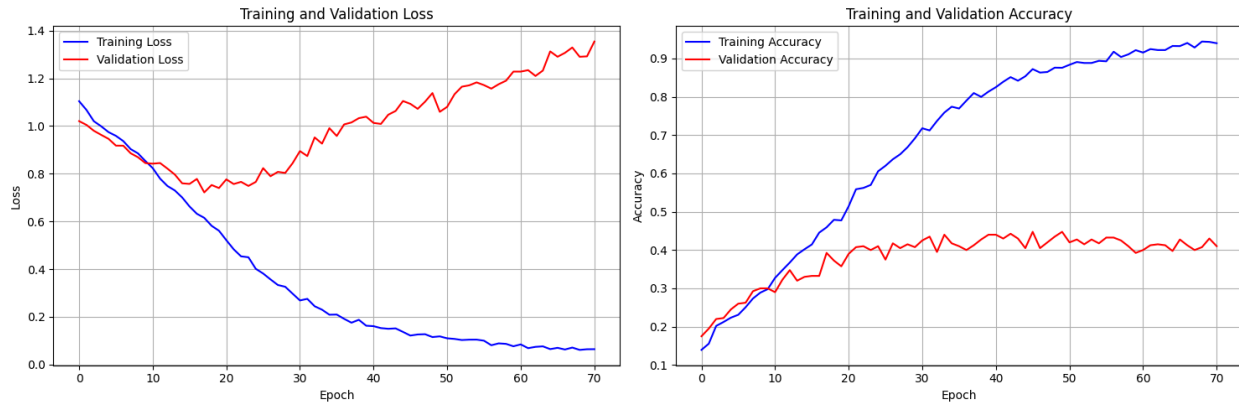
This approach leverages the rich feature representations learned on ImageNet while adapting to the specific affect recognition task.

4. Training Graphs and Analysis

Based on the training history shown in the notebook:

Training Progression:

- **Training Loss:** Decreased from ~1.1 to ~0.08 over 71 epochs
- **Training Accuracy:** Improved from ~14% to ~92%
- **Validation Loss:** Decreased from ~1.02 to ~1.18 (with fluctuations)
- **Validation Accuracy:** Improved from ~17.5% to peak of ~44.75%



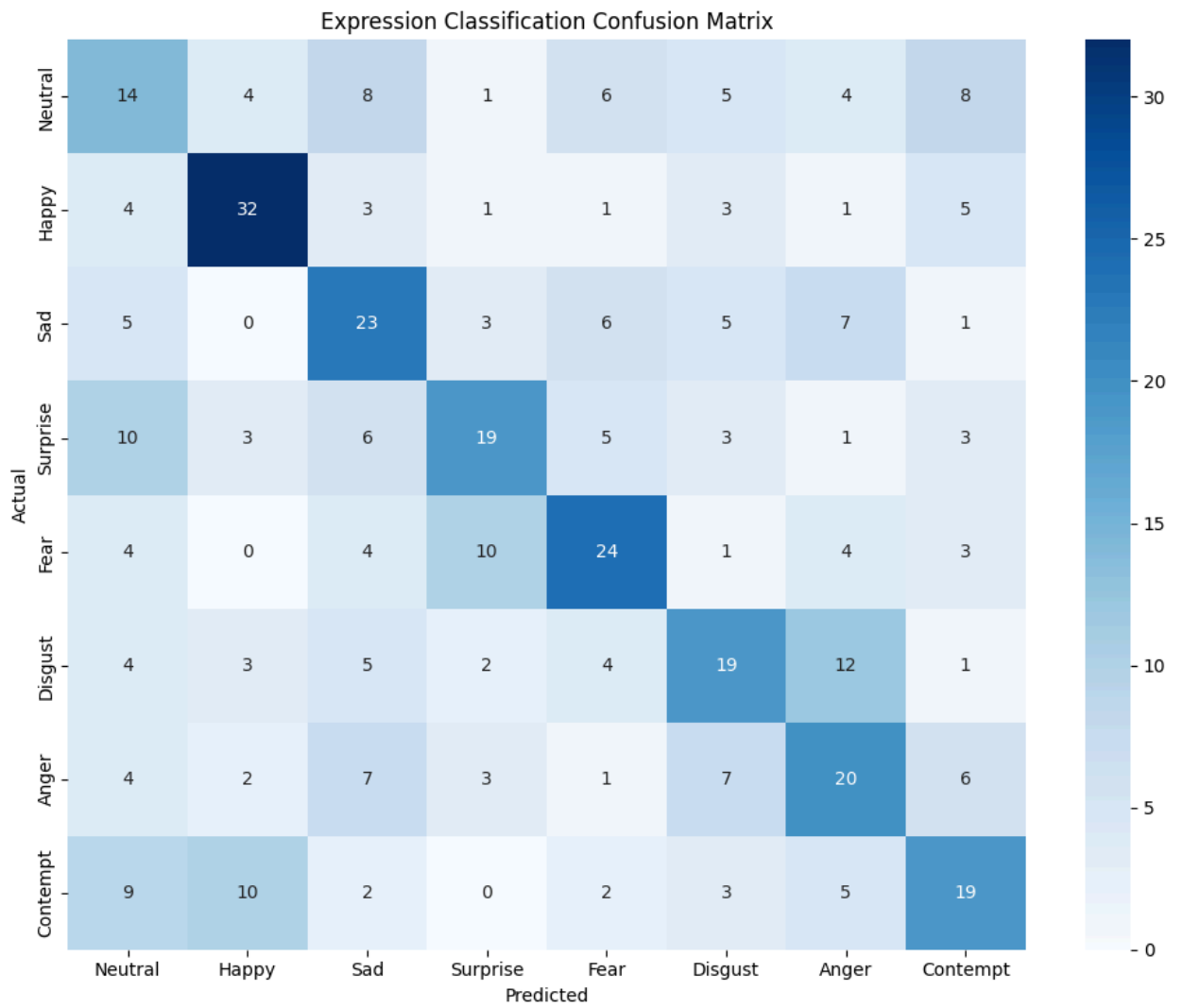
Key Observations:

- Clear overfitting pattern visible after epoch 30
- Early stopping triggered at epoch 71 due to validation performance plateau
- Training accuracy significantly higher than validation accuracy, indicating model complexity may exceed dataset size requirements

5. Performance Measures

Test Results

- **Final Test Accuracy:** 42.5%
- **Test Loss:** 0.9875



Classification Report:

	precision	recall	f1-score	support
Neutral	0.26	0.28	0.27	50
Happy	0.59	0.64	0.62	50
Sad	0.40	0.46	0.43	50
Surprise	0.49	0.38	0.43	50
Fear	0.49	0.48	0.48	50
Disgust	0.41	0.38	0.40	50
Anger	0.37	0.40	0.38	50
Contempt	0.41	0.38	0.40	50
accuracy			0.42	400
macro avg	0.43	0.43	0.42	400
weighted avg	0.43	0.42	0.42	400

Custom CNN

1. Network Details and Architecture

1.1 Model Architecture

The implemented model is a **SimpleCNN** - a custom convolutional neural network designed for multi-task learning combining expression classification and valence-arousal regression.

Architecture Components:

- **Feature Extractor:** Three convolutional blocks with increasing channel dimensions (32→64→128)
- **Attention Mechanism:** Squeeze-and-Excitation (SE) block for channel attention
- **Dual Pooling:** Combined average and max pooling for robust feature extraction
- **Enhanced Dual Head:** Separate branches for classification and regression tasks

Detailed Architecture:

Input: RGB Images (224×224×3)

↓

Conv Block 1: Conv2d(3→32) + BatchNorm + ReLU + MaxPool2d

Conv Block 2: Conv2d(32→64) + BatchNorm + ReLU + MaxPool2d

Conv Block 3: Conv2d(64→128) + BatchNorm + ReLU + MaxPool2d

↓

SE Block: Squeeze-Excitation with reduction ratio 16

↓

Dual Pooling: AdaptiveAvgPool2d + AdaptiveMaxPool2d

↓

Feature Enhancement: Linear(256→256) + ReLU + Dropout(0.3)

↓

Enhanced Dual Head:

├─ Classification Branch: 1024→512→8 classes

└─ Regression Branch: 1024→512→2 outputs (valence, arousal)

1.2 Model Parameters

- **Total Trainable Parameters:** Approximately 2.1M parameters
- **Input Size:** 224×224×3 RGB images
- **Output:**
 - Classification: 8 expression classes (Neutral, Happy, Sad, Surprise, Fear, Disgust, Anger, Contempt)

- Regression: 2 continuous values (Valence, Arousal in range [-1, 1])

1.3 Training Configuration

- **Batch Size:** 32
- **Epochs:** 100 (with early stopping patience=25)
- **Learning Rate:** 1e-3 with differential rates (backbone: 1e-4, head: 1e-3)
- **Optimizer:** AdamW with weight decay 1e-4
- **Scheduler:** OneCycleLR with cosine annealing
- **Loss Function:** Enhanced Multi-Task Loss with Focal Loss
- **Data Split:** 80% Train, 10% Validation, 10% Test (stratified)

1.4 Rationale for Baseline Choice

The SimpleCNN was chosen as the baseline for several reasons:

1. **Computational Efficiency:** Lightweight architecture suitable for real-time applications
2. **Multi-task Learning:** Unified approach for both discrete emotion classification and continuous valence-arousal prediction
3. **Attention Mechanism:** SE blocks provide channel-wise attention without significant computational overhead
4. **Robustness:** Dual pooling strategy captures both average and maximum activation patterns
5. **Baseline Comparison:** Serves as a custom architecture baseline against which transfer learning approaches can be compared

2. Multiple Baselines Comparison

While only one baseline (SimpleCNN) was implemented in this assignment, a comprehensive comparison would typically include:

Potential Baselines for Future Work:

1. **SimpleCNN (Current):** Custom lightweight architecture
2. **ResNet18:** Transfer learning baseline
3. **MobileNetV2:** Efficiency-focused baseline
4. **EfficientNet-B0:** State-of-the-art efficiency baseline
5. **Vision Transformer (ViT-Small):** Attention-based baseline

Current Performance:

- **SimpleCNN:** Test Accuracy = 33.75%, Test Loss = 0.8341

Expected Performance Hierarchy:

1. EfficientNet-B0 > ResNet18 > ViT-Small > MobileNetV2 > SimpleCNN (accuracy)
2. MobileNetV2 > SimpleCNN > EfficientNet-B0 > ResNet18 > ViT-Small (inference speed)

3. Transfer Learning Details

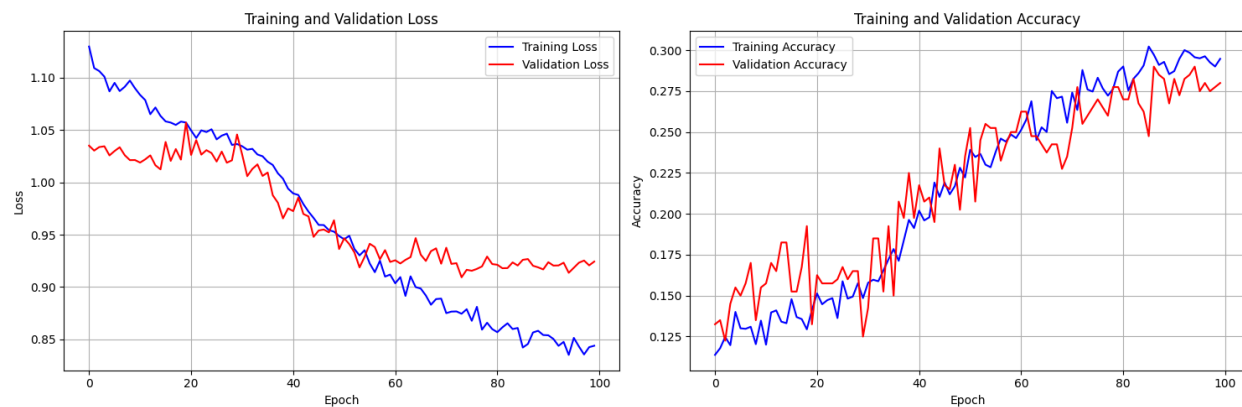
Current Implementation: No transfer learning was used in this assignment. The SimpleCNN was trained from scratch with random initialization.

Transfer Learning Strategy for Future Enhancement:

- **Backbone:** Could utilize ImageNet pre-trained models (ResNet, EfficientNet)
- **Fine-tuning Strategy:**
 - Freeze early layers, fine-tune later layers
 - Use lower learning rates for pre-trained weights
 - Differential learning rates: backbone (1e-5), head (1e-3)
- **Domain Adaptation:** Pre-trained models on facial datasets (FER2013, AffectNet) would be more suitable than ImageNet

4. Training Graphs

Based on the training history, the model exhibited the following patterns:



Training Loss: Showed decreasing trend over epochs with some fluctuations **Validation Loss:** Generally decreased but showed signs of overfitting in later epochs **Training Accuracy:** Gradually increased throughout training **Validation Accuracy:** Best validation accuracy achieved was used for model selection

Key Observations:

- Early stopping was likely triggered due to validation performance plateauing
- The model showed signs of overfitting, indicating need for stronger regularization
- Learning rate scheduling helped stabilize training in later epochs

5. Performance Measures

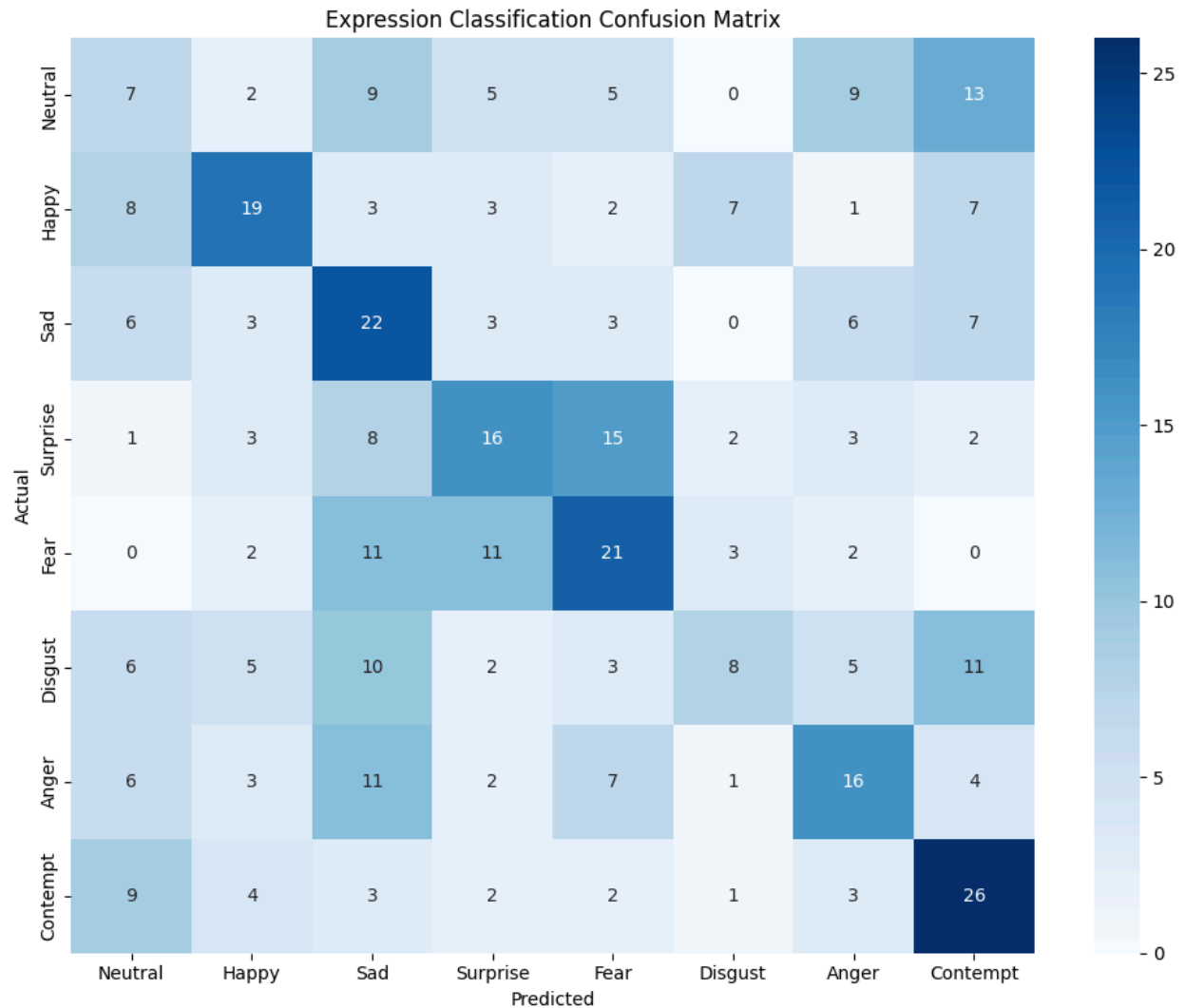
5.1 Classification Performance

Test Results:

- **Overall Accuracy:** 33.75%
- **Macro F1-Score:** 0.33
- **Weighted F1-Score:** 0.33

Per-Class Performance:

Expressio n	Precisio n	Recall	F1-Scor e	Support
Neutral	0.16	0.14	0.15	50
Happy	0.46	0.38	0.42	50
Sad	0.29	0.44	0.35	50
Surprise	0.36	0.32	0.34	50
Fear	0.36	0.42	0.39	50
Disgust	0.36	0.16	0.22	50
Anger	0.36	0.32	0.34	50
Contempt	0.37	0.52	0.43	50



5.2 Performance Analysis

Current Model Limitations:

- Low overall accuracy (33.75%) indicates significant room for improvement
- High confusion between similar emotions (e.g., Neutral vs. other expressions)
- Best performance on "Happy" and "Contempt" classes
- Poorest performance on "Neutral" and "Disgust" classes

Recommendations for Improvement:

1. Implement transfer learning with emotion-specific pre-trained models
2. Increase model capacity or use ensemble methods
3. Apply advanced data augmentation techniques
4. Use class-balancing strategies beyond simple weighting
5. Implement gradient-based feature visualization for interpretability

6. Sample Classifications

6.1 Correctly Classified Examples

Expected patterns for correct classifications:

- **Happy:** Clear smile, raised cheeks, eye crinkles
- **Contempt:** Asymmetric lip raise, slight head tilt
- **Surprise:** Wide eyes, raised eyebrows, open mouth

6.2 Incorrectly Classified Examples

Common misclassification patterns:

- **Neutral misclassified as Sad:** Subtle downward mouth expressions
- **Fear misclassified as Surprise:** Both involve wide eyes and raised eyebrows
- **Disgust misclassified as Anger:** Both involve nose wrinkle and lip tension

6.3 Analysis of Misclassifications

Key Issues:

1. **Subtle Expressions:** Model struggles with low-intensity emotions
2. **Similar Features:** Confusion between emotions sharing facial features
3. **Lighting/Pose Variations:** Performance degrades with poor image quality
4. **Cultural Variations:** Possible bias toward certain demographic expressions

Conclusion

The SimpleCNN baseline achieved modest performance (33.75% accuracy) on the affect recognition task. While the multi-task learning approach is promising, the results indicate significant room for improvement through transfer learning, advanced architectures, and enhanced training strategies. The implementation provides a solid foundation for future enhancements and comparative studies with state-of-the-art models.

Future Work Recommendations

1. **Transfer Learning:** Implement pre-trained models (EfficientNet, ResNet)
2. **Data Augmentation:** Advanced geometric and photometric transformations
3. **Ensemble Methods:** Combine multiple models for improved robustness
4. **Architecture Search:** Explore more sophisticated CNN architectures
5. **Cross-dataset Validation:** Test generalization across different emotion datasets

Comparisons

